Loss design

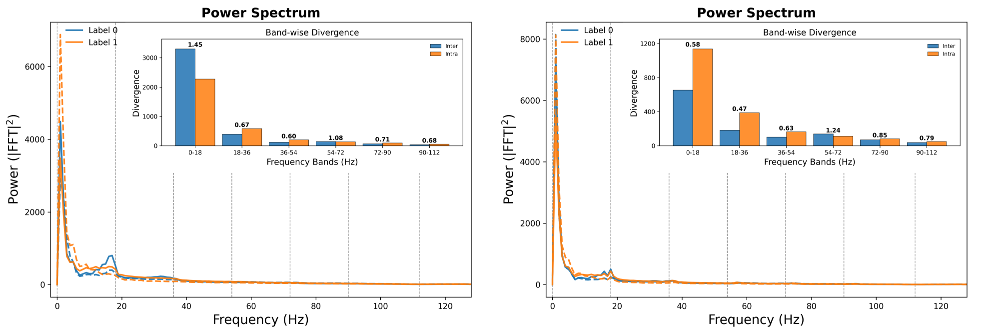
$$\mathcal{P} = \{(i,j) \mid y_i = y_j,\ s_i \neq s_j\}$$

$$\mathcal{L}_{\text{band-align}} = \sum_{k=1}^{K} \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left\| z_i^{(k)} - z_j^{(k)} \right\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{band}} \mathcal{L}_{\text{band-align}}$$

$z_i^{(k)}$ denotes the representation of sample i in the k -th band, and the loss is computed over all label-consistent but subject-discrepant pairs within each mini-batch($\mathcal{P}$), involving all eligible samples.

Result



(a) w/o $\mathcal{L}_{\text{band-align}}$ (b)w/ $\mathcal{L}_{\text{band-align}}$

Analysis

This figure shows that the model without $\mathcal{L}_{\text{band-align}}$ achieves more favorable spectral separation than the variant with the loss. In panel (a), the two classes exhibit clearer differences, particularly in the low-frequency region and around several secondary peaks, indicating stronger discriminative spectral structures. The band-wise divergence plot also suggests larger inter-class discrepancy in several key bands, which is beneficial for class separation. In contrast, after introducing $\mathcal{L}_{\text{band-align}}$ in panel (b), the spectral profiles of the two classes become more similar across multiple frequency bands. Although this loss is intended to encourage frequency-wise alignment, the visualization suggests that it may over-constrain the representations and reduce class-discriminative information.

Conclusion

Explicit structural alignment across components can harm cross-subject generalization and lead to degraded performance.